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Analysis of shockwave formation in glass welding by ultra-short pulses

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Abstract

Glass welding by ultra-short pulsed lasers is temporally a highly dynamic process. These dynamics have a high impact on the shape and the thermodynamic properties of the molten zone as well as the plasma. In this work we analyze the local distribution of laser absorption in the melt by shockwave observation and the evolution of the subsequent pressure wave generated during the process. The pressure waves were observed using a pump-probe setup with a high-speed video camera. Furthermore we show a dramatic decrease of pressure waves propagation velocity inside the molten zone as an explicit indicator of the fluid state of the molten zone.

Keywords: glass welding; glass processing; pump probe; ultra-fast lasers; sensing; image processing; ultra-short pulses;

1. Introduction

Glass welding by ultra-short pulsed (USP) lasers consists of an interplay of several effects on different timescales leading to a highly dynamic process. The process is started by nonlinear absorption [1,2] (e.g. multi-photon ionization) of the cold (room temperature) glass material which is fully transparent for the laser wavelength. Depending on the properties of the focusing optics and pulse duration also self-focusing effects may influence the shape of the focal spot generating the free electrons [3]. Depending on the pulse duration and energy of the pulse the generated electrons (initially by purely nonlinear effects) can be accompanied by avalanche ionization [4] (still within the same pulse) which can increase dramatically the absorptivity of the irradiated region and can be followed thus by undesired crack formation. The generated free electrons will transfer their kinetic energy subsequently to the glass lattice and heat the glass material. If the next pulse arrives "early" and overlapped enough with respect to the first pulse (by choosing a suitable pulse repetition rate and feed speed) the temperature of the glass will not returned back to room temperature before the arrival of the next pulse leading to heat accumulation in the focal region with each incoming pulse [5] until a quasi-static state is reached. The maximum achievable temperatures can lie in the range of few thousand degrees [5]. Of course, even at comparably lower temperatures (several hundred °C) free electrons can be generated by thermal ionization. Since plasma, i.e. free electrons, is typically an outstanding absorber [6,7], the ionization process can take place far above the actual focus position as can be seen in Fig. 1 and was shown by simulations in [8]. Moreover, the plasma generated "upstream" of the incoming laser beam will shield the irradiation inside laser spot [9].

Nonetheless, laser absorption above the nominal focal spot by plasma is not a stable state and results thus in quasi-periodic plasma movement of plasma generated in the focal spot moving towards the incoming laser beam until the local

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temperature decreases too far for a sufficient number of thermally ionized seed electrons for avalanche ionization, as has been shown by experiments [8,10] and simulations [8].

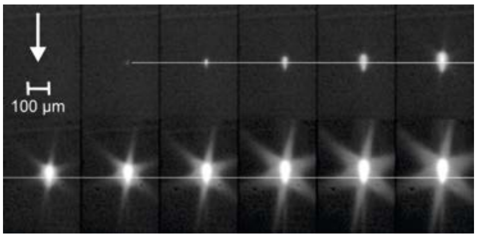

Fig. 1. 12 sequential images of plasma plume in fused silica bulk. The nominal focus position is indicated by the white line. The time delay between two adjacent images is 40 µs, the exposure time is 2 µs. The irradiation parameters are: NA 0.55, laser power 6 W, feed speed 20 mm/s, pulse repetition rate 100 kHz and 10 ps pulse duration. The laser beam propagation direction is indicated by the white arrow.

The simulation model considers interactions between the laser beam and the contributions of multiphoton, avalanche and thermal ionization processes. The simulations predict a strongly localized region of absorption inside the molten zone moving upstream towards the laser source. Concurrently, high-speed camera observations of the glass welding process show an apparent motion of one or more bright spots – assumed to be plasma – towards the laser source. However, while it is reasonable to assume that the experimentally observed bright spots correspond to the localized absorption regions in the simulations the hitherto conducted investigations cannot guarantee this since the bright spots could be caused by any number of possible effects and the simulation uses some assumptions and simplifications.

Therefore, the aim of this work is to show evidence that the bright spots in the high-speed movies of the welding zone and the localized absorption centers predicted by simulations are indeed the same. We do it here by showing that there exist pressure waves (PWs) initiated by shock waves generated inside the molten zone, which must originate from a tightly localized region. This implies that their place of origin must absorb the laser pulse significantly stronger than the rest of the molten zone.

2. Methods

Excerpts of relevant video sequences can be found at [11]. The experimental setup used to observe the PWs is described in detail in [10]. Still, for a self-consistent description of the experiment a brief summary of the setup is given: The laser, running at 1064 nm with 10 ps FWHM duration pulses, is split into a pump and a probe beam. The pump beam is focused by an aspheric lens with NA 0.51 into a 1 mm thick D263 borosilicate glass sample through a polished facet on its long side 200 µm below its surface (geometric distance) with 3.3 W average power after passing the lens. The probe beam is first frequency doubled, its diameter expanded, collimated and delayed in an adjustable delay line. Finally, the probe beam is launched through the pump interaction region (focal spot and molten zone) orthogonally to the pump beam and the moving direction of the sample. The probe beam transmitted through the interaction region is filtered by a laser line pass filter at 532 nm and finally imaged by a high-speed camera synchronized to the laser repetition rate (450 kHz) taking every second picture (camera frame rate 225 kHz). At the used magnification the nominal camera resolution lies at 2.13 µm/pixel while the optical resolution is limited to approximately 3 µm.

The visibility of the PWs is sufficient for unaided observation, see Fig. 2 (a), only if the pulses generate plasma in solid, i.e. not yet molten glass. Nonetheless, careful frame by frame observation of the high-speed videos has revealed that PWs can still be generated inside the melt. Thus an image processing algorithm was developed to enhance the display of the PWs, compare Fig. 2 (b) and (c):

(1): The brightness of each image is corrected by calculating the necessary contrast change from its average in a selected image region with constant features and applying the contrast change to the whole image. **(2)**: To enhance the PW for a given image a short image sequence of 25 images before and after (51 in total) this image is used to generate an averaged image. The image in question is then subtracted from this average. Near the beginning and end of the whole image sequence fewer images are taken for averaging. **(3)**: The contrast of the resulting image is strongly enhanced and subsequently filtered by a curvature flow filter [12].

While the above PW enhancement method provides sufficient contrast to observe the dynamic behavior of the PWs, further processing is required for quantification. Unfortunately, a detailed description of the necessary processing steps is out of scope for this paper's page limit. Suffice it to say that a rotating kernel transformation [13] was used to further suppress spurious noise and enhance circularly shaped contrast edges. A combination of edge detectors acting on different pixel scales [14] and ridge filters [15] with subsequent binarization was used to achieve a sufficiently adequate and stable representation of the PW structures, (Fig. 2 (e)). From these structures a convex hull was derived for which its bounding disk's point of origin and radius was calculated, (Fig. 2 (f)). While this method works well in case the size of the molten pool is small enough, the method breaks down if the molten pool becomes too large.

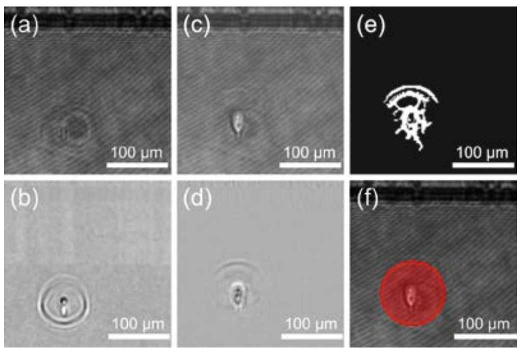

Fig. 2. (a) first image of a high-speed image sequence at feed speed 20 mm/s, probe delay 8.3 ns. (c) PW formation inside the molten pool after the first 76 pulses. (b) and (d) application of the PW enhancing algorithm on images (a) and (c). (e) binary representation of PWs. (f) bounding disk determined for PW representation structure (e) overlaid in red on image (c).

This is due to the following factors: **(1):** The PW generation is far less efficient in the melt pool, since the melt's compressibility is much higher than the compressibility of the solid glass. **(2):** The strong refractive index changes inside the melt pool and the plasma's presence as a strong light source overlay the structure of the PW propagating through the melt pool. **(3):** The speed of sound is significantly lower inside the melt than in the bulk material as shown for alumino-silicate glass in [16]. Albeit D263 is a borosilicate glass it is chemically similar to alumino-silicates so the results of [16] can be used for qualitative description. Thus, the spherical shapes of PWs become distorted due to the different propagation speeds inside and outside the melt pool, (Figs. 5, 6).

3. Results

The results of the PW propagation speed inside bulk material are presented first to establish the validity of the assumption that the PW's point of origin (PoO) can be estimated from its outer radius in case the melt pool is small enough.

3.1. Shockwave propagation speed

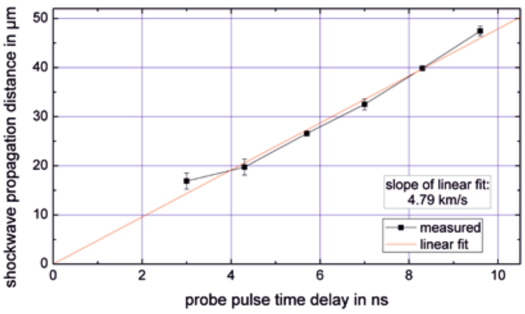

Fig. 3. PW propagation distance for the first impinging pulse observed at different delay times of the probe pulse.

The PW speed was determined by comparing the radius of the propagated PW for the first impinging pulse, i.e. in cold bulk material, at different probe delay times. The result is shown in Fig. 3 and gives a value of 4.79 km/s for the propagation speed. This is larger than the speed of a pressure wave propagating through isotropic compressible material by the Newton-Laplace equation with 3.2 km/s or a transversal sound wave at 3.47 km/s but still significantly slower than a longitudinal sound wave propagating at 5.71 km/s, both calculated from D263 glass's mechanical properties [17]. For the longest implemented probe delay of 9 ns in the experiment the highest and lowest possible speeds of sound would result in a travel length discrepancy 17.3 µm. At the available optical resolution this amounts up to ca. 8 pixels. This is indeed a value which can be observed for the width of the PW e.g. in Fig. 2 (b). Consequently, the optical resolution (3 µm) and the used probe delays are not high enough to be able to discriminate between contributions of longitudinal and transversal components as well as the potentially co-occuring diffraction effects on the refractive index change zone of the PW. The PW radii for probe delays at 3 ns were actually smaller than the region in which the initial plasma generation took place. The above described PW detection algorithm was dominated in these cases by the size of the plasma region (see Fig. 5 (c) and (d)). Thus the corresponding data were not used to estimate the PW speed by linear regression.

3.2. Shockwave's point of origin

The observation of a more or less spherical PW allows an extrapolation of its origin, thus allowing the determination of the localized absorptivity inside the melt. This approach is of course only feasible if the PW is larger than the molten zone. The above described algorithmic PW size determination imposes further limitations on the appearance of the PWs and the molten pool which is why in the following numeric analysis not all high-speed camera exposures were used. Furthermore, for some frames the algorithm gave incorrect values for the PW radius and thus also its origin. These points are not displayed in the analysis. Nonetheless, for the cases inaccessible to numeric evaluation qualitative behavior is shown.

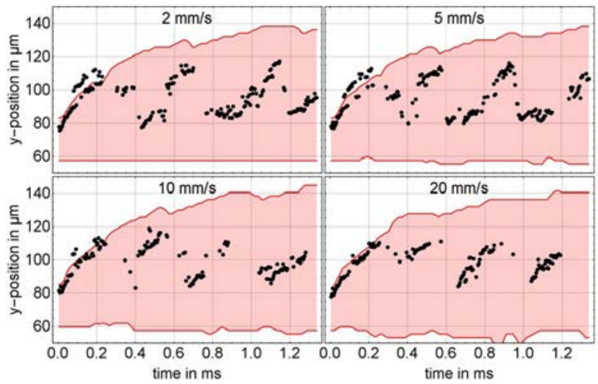

Fig. 4. Vertical position of the lower and upper boundaries of the melt pool in red and the determined vertical position of the PW's point of origin in black for different feed speeds at a probe time delay of 9.6 ns.

As can be seen in Fig. 4 during the initial stage with each newly arriving pulse, the PWs are generated at more elevated positions. Interestingly, the PW's PoO move synchronously with the growth of the upper boundary of the developing melt pool. (The laser direction is here the same as indicated in Fig. 1.) The initial growth rate of this can be estimated to 0.1 m/s on average for all investigated feed speeds (compare [8]). While the horizontal position of the PW's PoO changes over time too, the changes are not as pronounced as in the vertical axis, so we do not consider them in this work due to space limitations. However, the upward vertical movement of the PoOs slows first and then stops at a height of about 120 µm. This seems to be the limit beyond which avalanche ionization becomes too ineffective. In Fig. 4 the first PoO change occurs around 0.4 ms but at higher feed speeds (not shown here) this happens earlier. After this the PWs are generated again at the lower side of the melt pool near the nominal focal spot position, while the melt pool size continues to grow. Consequently, the laser absorption must be concentrated during this stage at the bottom of the melt while the regions above this position which were previously highly absorbing are no longer strong absorbers. This means that the absorption process of laser pulse must indeed be localized while being highly dynamic at the same time.

A further interesting aspect is the size of the molten pool for the very first few pulses, since its size is about 20 µm. Strictly speaking, it is not clear if this can be assumed to be a molten pool, because the material is likely to be still too cold to realize melting. However, the image evaluation algorithm is able to detect this zone due its brightness compared with the image background. It is likely that this is due to the recombination of the initial plasma generated by multi-photon and avalanche ionization (avalanche ionization within a single pulse). Nonetheless the PW's PoO lies near the top of this plasma zone. This is plausible because the top side of the existing plasma will act as a strong absorber thereby shielding the lower part of the initial plasma and thus preventing laser absorption and PW generation in the deeper parts.

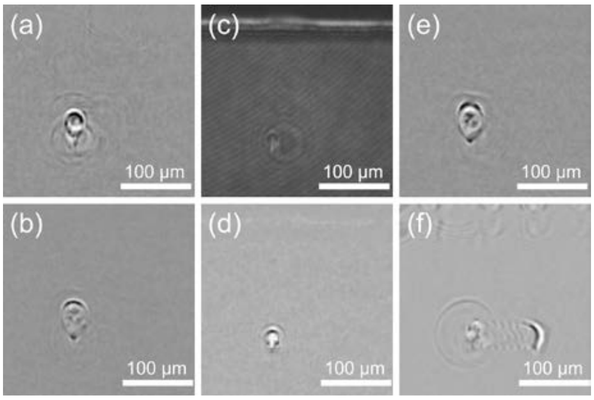

Fig. 5. (a) Simultaneous generation of PWs in one molten pool. (b) weak PW unsuitable for automated detection. (c) and (d) original image and PW enhanced image for at a small probe pulse delay time – the actual PW has not propagated far enough and is overlaid by the plasma region. (e) PW distortion due to different propagation speeds in and outside the molten pool. (f) Overlay of PW due to refractive modifications of the processed glass. The processing parameters are as follows: (a) feed speed 10 mm/s, 7 ns probe delay, 153rd pulse; (b) and (e) feed speed 2 mm/s, 7 ns probe delay, 508th and 530th pulse; (c) and (d) feed speed 2 mm/s, 3 ns probe delay, 6th pulse.

The lack of measurement points for certain time positions in Fig. 4 is however not an indicator that there is no PW. Sometimes the PWs are generated in parallel at the top as well as bottom of the molten zone. In such case the algorithm fails to detect the PW's PoO properly. The same applies for cases when the PWs are very weak or overlaid by other structures or not circularly shaped due to a slow down inside the molten pool (see Fig. 6). All cases are shown in Fig. 5.

3.3. Shockwave propagation speed in molten zone

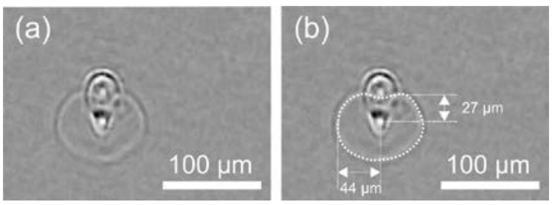

Fig. 6. (a) PW deformation due to propagation through molten zone. Feed speed 5 mm/s, probe delay 9.6 ns, effect of the 471st pulse irradiating the sample. (b) manually sketched shape of the PW in (a) with manual size measurements.

Fig. 6 shows the shape of a PW in the presence of relevant amount of melt. The shape of the PW deviates clearly from a circular shape as seen in Fig. 2 (a) and (b). The PW propagation speed inside the molten zone is estimated to be 2.8 km/s, just about 60% of the speed determined for cold bulk material, a very similar decrease as observed in [16]. Unfortunately, the distorted shape of the PW in these cases makes an automated analysis of its origin and propagation speed very difficult, especially when additional structures such as the molten zone or the plasma radiation overlay the PW features.

Comparison of the PW sides propagating perpendicularly to the long axis of the melt pool shows with 44 µm at 9.6 ns a significantly smaller propagation distance than initial pulse measurements in Fig. 3 resulting in an average propagation speed of 4.58 km/s. This is likely due to partial propagation of the shockwave through the melt pool with the lower speed.

4. Conclusions

The experiments show that the PW generation is highly localized and that the PW's PoO depends on the internal dynamics of the melt pool. Because of the PW's existence it is further plausible that a significant amount of the laser pulse's energy is absorbed during the PW generation.

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